

Mahi Singh

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering — CGPA: **9.20/10.00**

Bhopal, India

Aug 2023 – May 2027

Mercy Memorial School

ISC (Class XII): **89.75%** | ICSE (Class X): **95.40%**

Uttar Pradesh, India

2023 & 2021

Technical Skills

Languages: Java, Python, JavaScript, SQL

AI & Machine Learning: TensorFlow, PyTorch, YOLOv11, Computer Vision, Deep Learning

Web & Databases: React, Next.js, FastAPI, PostgreSQL, MongoDB

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook

Experience

AI Engineer Intern

Mar 2025 – Sep 2025

AmasQIS.ai

Oman (Remote)

- Engineered a real-time PPE compliance detection system using Python, TensorFlow, and Faster R-CNN to process live multi-camera factory feeds, achieving **90+ precision%** on the validation set.
- Developed an end-to-end object detection pipeline to classify safety violations (helmets, vests, gloves) across concurrent camera streams for active factory floor monitoring.
- Optimized model inference for low-latency production deployment; collaborated with cross-functional engineering teams on iterative model improvements through structured optimization cycles.

Projects

JeevanTrack — AI Health Timeline Platform | *Next.js, FastAPI, Supabase, Gemini 2.5, TypeScript* [GitHub](#)

- Built an AI-powered health record platform that digitizes, summarizes, and visualizes patient medical history into a unified timeline using Gemini Vision, Next.js, FastAPI, and Supabase.
- Integrated Google Gemini 2.5 Flash via Vision API to automate medical report text extraction, reducing manual data entry by **80%** and rendering concise 1-page clinical summaries in under **30 seconds**.
- Designed a resilient multi-tier fallback system (local storage → rules-based matching → manual flow) with offline-first caching and cloud sync, ensuring **100% reliability** and zero data fragmentation across **5+ core features**.

knowYourFood — GenAI Ingredient Analyzer | *React, FastAPI, Python, Gemini 2.0, PostgreSQL* [GitHub](#)

- Developed a GenAI-powered food label analyzer that uses Google Gemini to explain ingredient safety, identify potentially harmful additives, and generate consumer-friendly health insights in real time.
- Architected the platform with a React frontend and asynchronous FastAPI backend, achieving a **13 second** average response time across **30+ product analyses**.
- Designed a secure REST API with structured prompt engineering, PostgreSQL authentication, and strict CORS security policies; implemented responsible AI output validation for safe, accurate responses.

KharpatNaashak — AI Weed Detection System | *YOLOv11, PyTorch, OpenCV, Gradio, Python* [GitHub](#)

- Developed a computer vision system for automated weed detection in soybean fields using YOLOv11, enabling precise identification of crop weeds from field images.
- Trained a YOLOv11n model identifying 16 weed species, achieving **60.7% mAP@0.5**, **67.1% precision**, and 52.2% recall on the **MH-Weed16 dataset (6,656 images)**.
- Built a bilingual Gradio web interface with video upload, density heatmaps, and a herbicide cost-savings calculator demonstrating a **60–70%** chemical reduction through localized herbicide targeting.

Leadership & Volunteer

Technical & Community Volunteer

Jul 2023 – May 2025

eVidyaloka · Pahal Foundation · Pledge A Smile

- Taught mathematics for 2 years as a volunteer with eVidyaloka to 50+ students in rural Madhya Pradesh, improving assessment scores by 30%.
- Tutored construction workers' children of all ages through a volunteer teaching initiative (Pahal Foundation), helping some transition into formal schooling; also ran poster design and fundraising social campaigns for Pledge A Smile.